

Completeness $\wedge$ Closed-graph continuity $\wedge$ Betweenness $\Rightarrow$ Transitivity

Conjecture — human; produced by ZG.

0.1 Premises

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.
- **Betweenness.** For all $p, q \in \Delta(X)$ and $\lambda \in (0,1)$: if $p \succ q$ then $p \succ \lambda p + (1-\lambda)q \succ q$, and if $p \sim q$ then $p \sim \lambda p + (1-\lambda)q$.

0.2 Conclusion

- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.

0.3 Notes

Sample conjecture.

Record: topics/decision-theory/results/betweenness-continuity-imply-transitivity.yaml